

Modelling

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Computational analyses of the biological systems

- **System structure**
 - the genes interactions network
 - the biochemical pathways
 - the mechanisms by which these interactions affect the chemo-physical properties of the intra and inter cellular apparatus
- **The Dynamics** describes the behaviour of the system in time, under different conditions by means of:
 - metabolism analysis and environmental adaptation
 - methods of dynamics analysis (e.g. phase space, bifurcations, periodicities, ...)
 - identification of the pivotal mechanisms underlying specific behaviours

Complex Systems

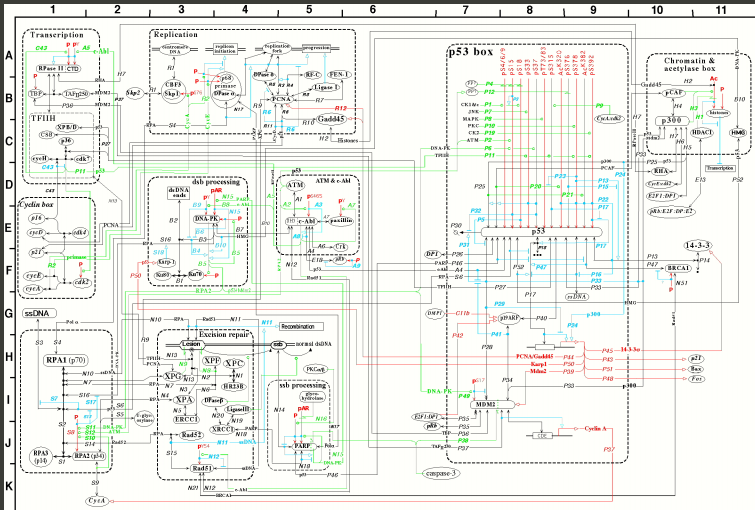


Figure 6B: The p53-Mdm2 and DNA repair regulatory network (version 2p - May 19, 1999)

Cellular Networks

According to the functionality ($\neq$ players) it is possible to distinguish two networks:

- **metabolic** network - mass/energy flux

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- **metabolic** network - mass/energy flux
- **signalling** network - information flux

Metabolic network

Composed by:

- metabolites
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 - fatty acids
 - amino acids

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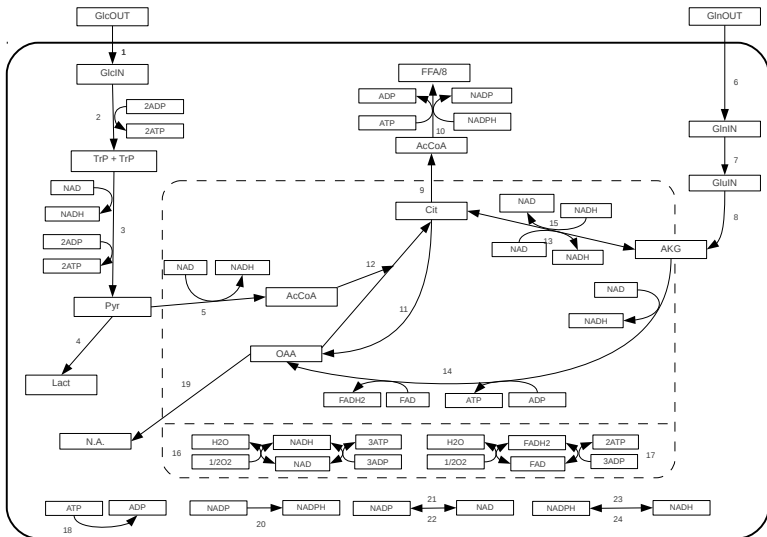
Metabolic network

Composed by:

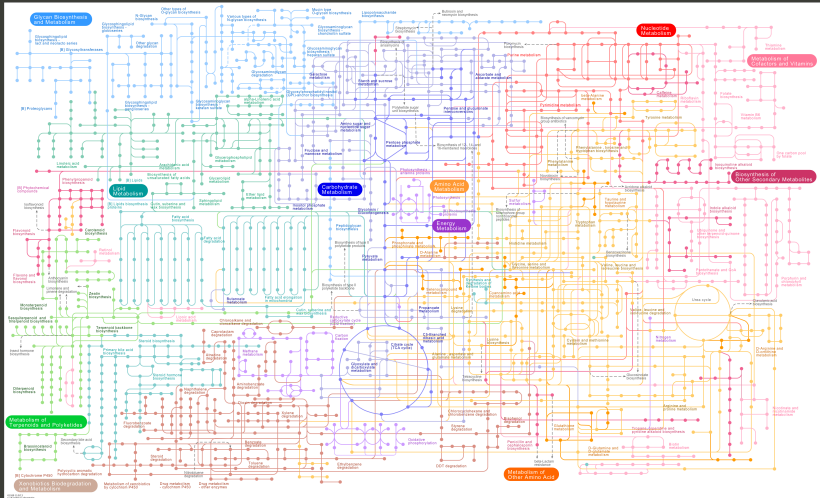
- **metabolites**
 - glucose
 - fatty acids
 - amino acids
- **enzymes** (proteins)

The cell exploits metabolites, through enzymes, to “produce” and “consume” the energy and the mass required to synthesize cellular components.

Metabolic network



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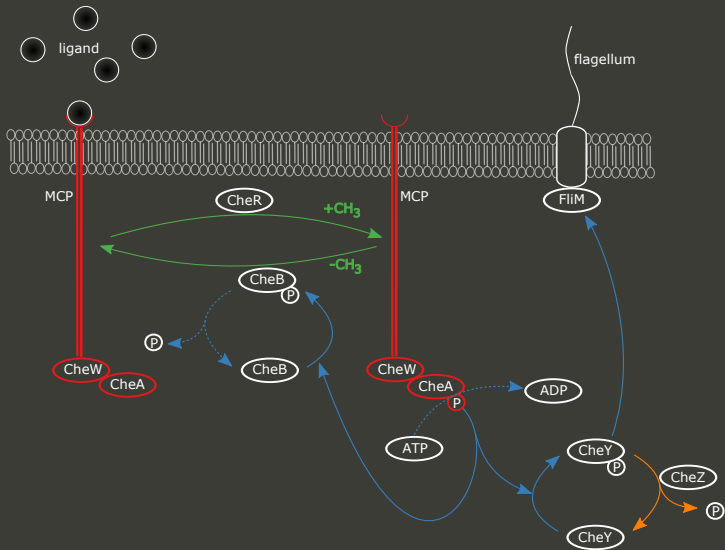
Signalling network

Composed by:

- **information**
 - DNA (genes)
 - mRNA
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It is responsible for the coding/decoding of the inter/intra -cellular information. It allows cellular differentiation, stimuli reactions, growth, ...

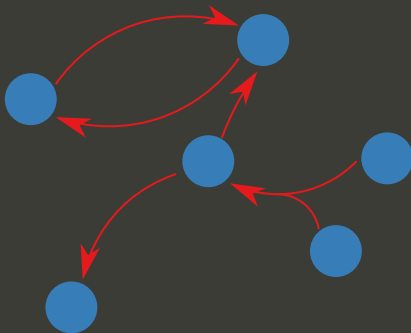
Bacterial Chemotaxis



Cellular Networks

The constitutive elements to build a model of a cellular network are:

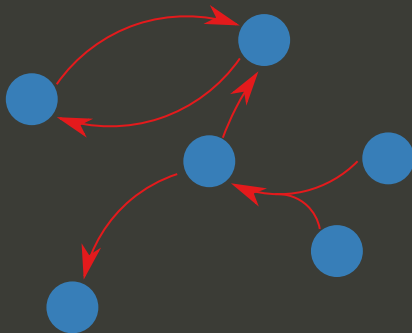
- the molecular *players*
- the list of the *interactions* among the players



Networks

The constitutive elements to build a model of a **network** are:

- the **players**
- the list of the **interactions** among the players

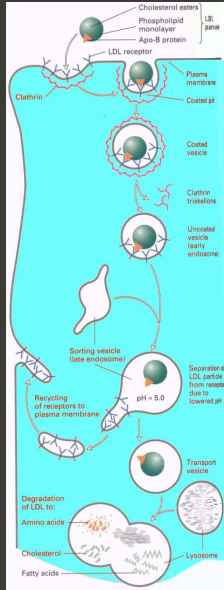


Modelling approaches

It is possible to identify **three abstraction levels** to represent information and to model a system:

- graphical
- qualitative
- mechanistic

Graphical models



They describe each biological component through drawings connected by arrows in case of interaction.

Graphical models

Molecular interaction map of the mammalian cell cycle control and DNA repair systems- Khon K.W.

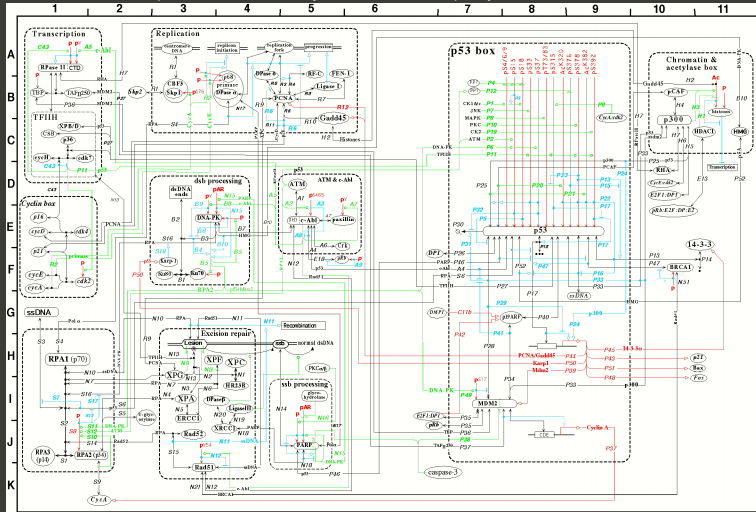
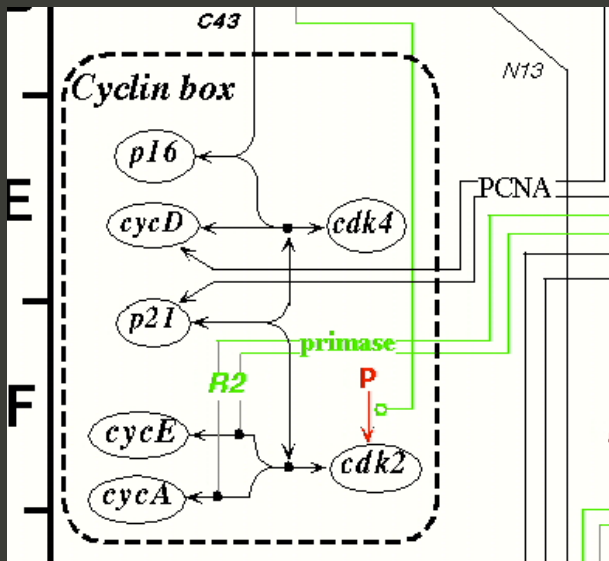


Figure 6B: The p53-Mdm2 and DNA repair regulatory network (version 2p - May 19, 1999)

Graphical models



Graphical models

- pros
 - they allow to synthesize and exemplify a huge amount of information
 - they facilitate the hypotheses building on the network functioning
 - they highlight issues on the network understanding
- cons
 - even the most detailed model is built on top of simplifications and lacking of information
 - they are static

Qualitative models

They ranges from simple graphs to logical or statistical models.

- for a first simulation of complex networks it is possible to use:
 - systems based on *Boolean/fuzzy logic*
 - *rule-based* systems (generative)
- if it is available information on the perturbations response or on time series it is possible to exploit *statistical inference*
- if the focus is on the topology or on the network dynamics it is possible to exploit
 - Boolean networks
 - linear networks
 - Bayesian networks

Qualitative models

- **pros** they can be
 - analyzed in fully automatic manner
 - generated even in case of lacking information
- **cons**
 - they require a massive data interpretation increasing the probability to hard coding the desired system response
 - not strictly causal (built looking for some ad hoc behaviour)

Mechanistic models

They are highly specific models describing each interaction in the system relying on chemo-physical laws (causality).

usually two kinds of processes are involved:

- **deterministic** - the knowledge on the system states up to time t allows the *unique and determined evolution* of the system at time $t + \Delta t$.

$$X(t + \Delta t) = X(t) + f(X(t), t) dt; \quad \frac{dX}{dt} = f(X, t)$$

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- **stochastic** - the knowledge of the system states up to time t allows *probabilistic evolution* of the system at time $t + \Delta t$

Mechanistic models

two possible description for the state of the system:

- continuous $\mathbb{R}$

the system's states belongs to *real numbers*

e.g. the reactants molecular *concentration*

Mechanistic models

two possible description for the state of the system:

- **continuous** $\mathbb{R}$
the system's states belongs to *real numbers*
e.g. the reactants molecular *concentration*
- **discrete** $\mathbb{N}$
the system's states belongs to *natural numbers*
e.g. the reactants copy numbers (individual molecules)

Mechanistic models

- pros
 - causal structure
 - the most easy to falsify
 - highly precise and detailed predictions
- cons
 - they require a huge amount of data
 - they require a very detailed knowledge of the network
 - high computational cost

Experimental Evidences of Noise in Biology

Many **experimental evidences** of stochasticity in living systems:

- **transcription and translation** [Abkowitz, 1996; Ozbudak, 2002]
- **mRNA production** is "quantal" [Hume, 2000] and in random pulses [Ross, 1994; Walters, 1995]
- the **protein production** occurs in short bursts and at random time intervals [Yarchuk, 1992; Chapon, 1982]
- in **the λ phage** the same starting conditions lead the system to two different kinds of evolutions (lysis/lysogeny) [Oppenheim, 2005 (review)]
- ...

Experimental Evidences of Noise in Biology

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Fig. 1. Intrinsic and extrinsic noise can be measured and distinguished with two genes (*cfp*, shown in green; *yfp*, shown in red) controlled by identical regulatory sequences. Cells with the same amount of each protein appear yellow, whereas cells expressing more of one fluorescent protein than the other appear red or green. (A) In the absence of intrinsic noise, the two fluorescent proteins fluctuate in a correlated fashion over time in a single cell (left). Thus, in a population, each cell will have the same amount of both proteins, although that amount will differ from cell to cell because of extrinsic noise (right). (B) Expression of the two genes may become uncorrelated in individual cells because of intrinsic noise (left), giving rise to a population in which some cells express more of one fluorescent protein than the other.

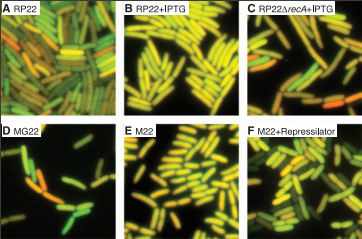
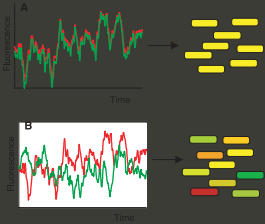


Fig. 2. Noise in *E. coli*. CFP and YFP fluorescence images were combined in the green and red channels, respectively. (A) In strain RP22, with promoters repressed by the wild-type *lacI* gene, red and green

Experimental Evidences of Noise in Biology

vt, and approved October 8, 2008 (received for review September 3, 2008)

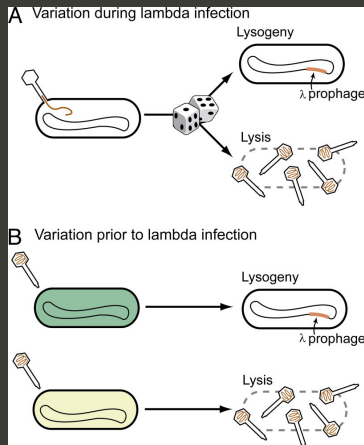
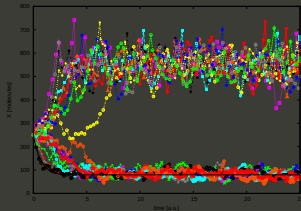


Fig. 1. Alternative models for cell fate selection in a population of genetically identical cells. (A) Variation during infection (for example, spontaneous chemical kinetic noise) leads to qualitative differences in cell fate. (B) Variation in the physical state of individual cells before infection predetermines cell fate.

St. Pierre, PNAS 105 (2008)

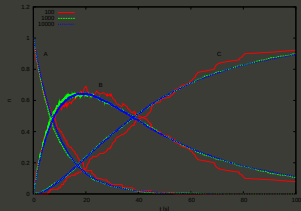
Experimental Evidences of Noise in Biology

- **intrinsic** noise - due to the inherent nature of the biochemical interactions



same initial conditions $\neq$ evolutions

- **extrinsic** noise - due to the external environmental conditions



$$\text{noise influence} \propto \frac{1}{\sqrt{n}}$$

Experimental Evidences of Noise in Biology

- Biological systems can be extremely **non-linear** and often exhibit **many steady states, bifurcations or chaotic behavior**
- Stochastic simulation is the probe to access the different evolutions

Deterministic Vs Stochastic Approaches

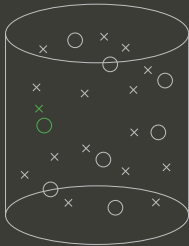
Chemical Kinetics - two perspectives:

Deterministic Formulation



- kinetic constants $K \equiv$ reaction "rates"
- molecular concentrations $[x] \equiv$ scalar function of time

Stochastic Formulation



- kinetic constants $K \equiv$ reaction "probabilities" rates
- the time evolution of the chemical system $\equiv$ a N -dim Markovian random walk in the molecular species space

Deterministic Vs Stochastic Approaches

The **deterministic approach** - Ordinary Differential Equations:

$$\begin{aligned}\frac{dx_1}{dt} &= f_1(x_1, \dots, x_n) \\ &\vdots \\ \frac{dx_n}{dt} &= f_n(x_1, \dots, x_n)\end{aligned}$$

- molecular and environmental interactions are described by means of **an equation for each molecular species x_i**
- it requires the **simultaneous solving** of all the equations
- it captures the **ensemble features** of the system (global)

Deterministic Vs Stochastic Approaches

The **stochastic approach**:

- molecular interactions are described by means of **probability distributions**
- the probability distributions are **dynamic**. They evolve according to the system state
- it exploits a **scattering perspective** of chemical reactions
- it captures **individuals behavior** of the molecules (local)

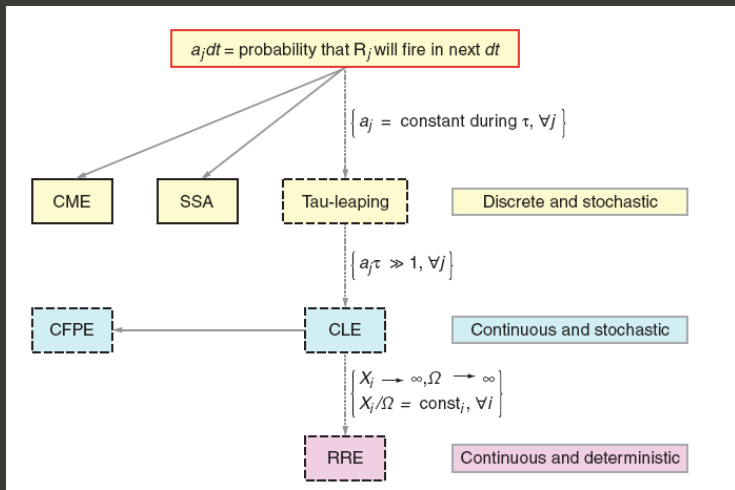
Deterministic Vs Stochastic Approaches

Statistical physics argumentation shows that the stochastic approach:

- is always **valid when deterministic is**
- may be **valid when ordinary deterministic is not**, i.e. in a nonlinear system in the neighborhood of a chemical instability
- fully accounts for inherent **statistical correlations** and **fluctuations** neglected by the deterministic
- never approximates infinitesimal time increments by dt but **uses finite time steps Δt**

Stochastic approach requires **HUGE computational time** for big number of molecules!

Modelling Regimes



Description Scale

